

Bits, Bytes & Units — Detailed Lecture Notes

JKSSB Computer Awareness

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Video lesson: [Bits, Bytes & KB–TB Explained](#)

3.1 The basic units of digital data

Digital information is represented using bits. Groups of bits form larger units, but the terms do not all describe the same kind of quantity. A bit is a single binary digit; a word is an architecture-dependent processor unit; a character is an encoded symbol.

Definition (Bit). *A binary digit that has one of two values, 0 or 1; the smallest unit of data in the usual computer-fundamentals description.*

Definition (Nibble). *A group of four bits.*

Definition (Byte). *A group of eight bits, equal to two nibbles.*

Definition (Word). *The natural data size a processor handles as a unit; its size depends on the processor architecture.*

Definition (Character). *A text symbol represented in digital storage by a character encoding; its storage size is not necessarily one byte.*

Unit	Size / basis	Meaning
Bit	1 binary digit	One of the two values 0 or 1
Nibble	4 bits	Half of an 8-bit byte
Byte	8 bits	Two nibbles; common unit for file/storage amounts
Word	Architecture-dependent	Natural processor data size, such as 32 or 64 bits
Character	Encoding-dependent	A text symbol; its encoded storage may take more than one byte

The bit-to-byte relationships are fixed in this lesson:

$$1 \text{ nibble} = 4 \text{ bits}, \quad 1 \text{ byte} = 8 \text{ bits} = 2 \text{ nibbles}.$$

The word size is not defined by this ladder. A processor may work with a natural word size such as 32 or 64 bits, but the term depends on the architecture and should not be treated as one universal number.

Example 3.1.1. *Write the byte 10110110 as two nibbles: 1011 and 0110. Each group contains four bits; together the two groups still contain eight bits, so the original value is one byte.*

3.2 Bits and bytes in names and data rates

Capitalization changes the unit. Lowercase **b** denotes a bit, while uppercase **B** denotes a byte. A byte contains eight bits. Therefore a rate reported in megabits per second is not numerically equal to a rate in megabytes per second.

Definition (Transfer rate). *The amount of data transferred in a stated time interval, commonly written in bits per second or bytes per second.*

Definition (Capacity). *The amount of data that a storage medium can hold.*

The slash in a rate such as Mbps means “per second.” It is not part of the data amount itself. Keep the numerical amount, whether it is bits or bytes, and the time unit visible while calculating. Under matching decimal prefixes,

$$1 \text{ byte} = 8 \text{ bits}, \quad 1 \text{ MB} = 8 \text{ Mb}.$$

Thus dividing a rate in Mb/s by eight gives MB/s, while multiplying a file size in MB by eight gives Mb. These conversions change the unit, not the underlying amount of data.

To convert a bit rate to a byte rate, divide by eight, provided the prefixes match and the question is asking for the ideal conversion. Protocol overhead, congestion, and other factors can reduce actual network throughput, so a nominal rate is not a promise of application-level speed.

Example 3.2.1. *A link is rated at 8 Mbps. Using matching decimal prefixes and ignoring overhead, convert bits to bytes:*

$$8 \text{ megabits/s} \div 8 = 1 \text{ megabyte/s}.$$

This is a transfer rate, not a storage capacity.

Example 3.2.2. *Convert an ideal rate of 24 Mbps to megabytes per second. The prefixes already match, so only the bit-to-byte conversion is needed:*

$$24 \text{ Mb/s} \div 8 = 3 \text{ MB/s}.$$

The answer is a rate because the time denominator remains. If the question asked how long a file takes to transfer, the file amount would first need to be expressed in the same unit as the rate, then divided by that rate.

Example 3.2.3. *A 4 MB file contains 32 megabits under matching decimal prefixes because $4 \text{ MB} \times 8 = 32 \text{ Mb}$. If transferred over an ideal 20 Mbps link, the simple lower-bound calculation is $32/20 = 1.6$ seconds. Actual transfer can take longer because the ideal calculation ignores protocol overhead and other delays.*

3.3 Capacity units and the 1024 convention

For the conventional JKSSB-style ladder used in this lesson, each step is a factor of 1024:

$$1 \text{ KB} = 1024 \text{ bytes}, \quad 1 \text{ MB} = 1024 \text{ KB}, \quad 1 \text{ GB} = 1024 \text{ MB}, \quad 1 \text{ TB} = 1024 \text{ GB}.$$

Thus 1 MB under this convention is $1024 \times 1024 = 1,048,576$ bytes. The exam-style abbreviations KB, MB, GB, and TB are often used with this traditional binary ladder. Modern technical standards distinguish decimal SI units (powers of 1000) from binary IEC units such as KiB (powers of 1024). If a question specifies decimal units or the IEC prefixes, use those definitions instead. Do not silently mix conventions.

Unit	Previous unit	Bytes under repeated 1024 convention
1 KB	1024 bytes	1,024 bytes
1 MB	1024 KB	1,048,576 bytes
1 GB	1024 MB	1,073,741,824 bytes
1 TB	1024 GB	1,099,511,627,776 bytes

To move from a larger unit to the next smaller unit, multiply by 1024. Repeat once for every step crossed. To move from a smaller unit to the next larger unit, divide by 1024. Writing the units beside the numbers helps prevent using the wrong direction.

Example 3.3.1. *Convert 3 KB to bytes under the 1024 convention:*

$$3 \text{ KB} \times 1024 \text{ bytes/KB} = 3072 \text{ bytes.}$$

The KB units cancel, leaving bytes.

Example 3.3.2. *Convert 2 MB to KB under the same convention:*

$$2 \text{ MB} \times 1024 \text{ KB/MB} = 2048 \text{ KB.}$$

This is one step down the capacity ladder, so multiply by 1024.

Example 3.3.3. *Convert 5 GB to MB using the same convention. GB is one step larger than MB, so multiply by 1024:*

$$5 \text{ GB} \times 1024 \text{ MB/GB} = 5120 \text{ MB.}$$

The units GB cancel, leaving MB. To convert 5120 MB back to GB, divide by 1024 and recover 5 GB.

3.4 Keep capacity separate from rate

Capacity answers “how much can be stored?” Transfer rate answers “how much is moved per unit of time?” MB is a byte-based amount; Mbps is a bit-based rate. The lowercase/uppercase distinction and the /s time component carry information. A drive with a larger capacity does not necessarily transfer data faster than a smaller drive.

Question	Capacity	Transfer rate
What does it measure?	Amount that can be held	Amount moved per unit of time
Common example	A 4 MB file or 1 TB drive	A 20 Mbps connection
Unit clue	No time denominator	Includes “per second” or /s

Example 3.4.1. *In the lesson’s opening comparison, 4 MB describes the file’s storage amount, while 20 Mbps describes the network’s transfer rate. The first has no time denominator; the second is measured per second, so they are different kinds of quantities.*

Before calculating, perform a quick unit check: if the final question asks for bytes, convert bits to bytes; if it asks for seconds, divide an amount of data by a compatible rate; if it asks for a storage amount, do not attach a time unit. A numerically correct calculation with mismatched units is not a correct answer.

Exercises

1. How many bits are in 7 bytes? Show the conversion.
2. How many bytes are in 5 KB under the 1024 convention?
3. Convert 24 Mbps to MB/s using matching decimal prefixes and ignoring overhead.
4. A device stores 4 GB. Is this a capacity or a transfer rate? What unit clue supports your answer?
5. A processor has a 64-bit word size. How many bytes are in one word?

Solutions

1. $7 \times 8 = 56$ bits.
2. $5 \times 1024 = 5120$ bytes.
3. $24/8 = 3$ MB/s before protocol overhead.
4. It is capacity: GB is a data amount, and there is no “per second” time component.
5. $64/8 = 8$ bytes per word.